WEARABLE TRACKER • ESP32 · NEO-6M · SIM800L · TP4056

Tracker Wiring

Pin-for-pin against `tracker/include/pins.h`. Power first — that is what kills these builds, and the failure looks like a firmware bug for days.

READ BEFORE CONNECTING ANYTHING

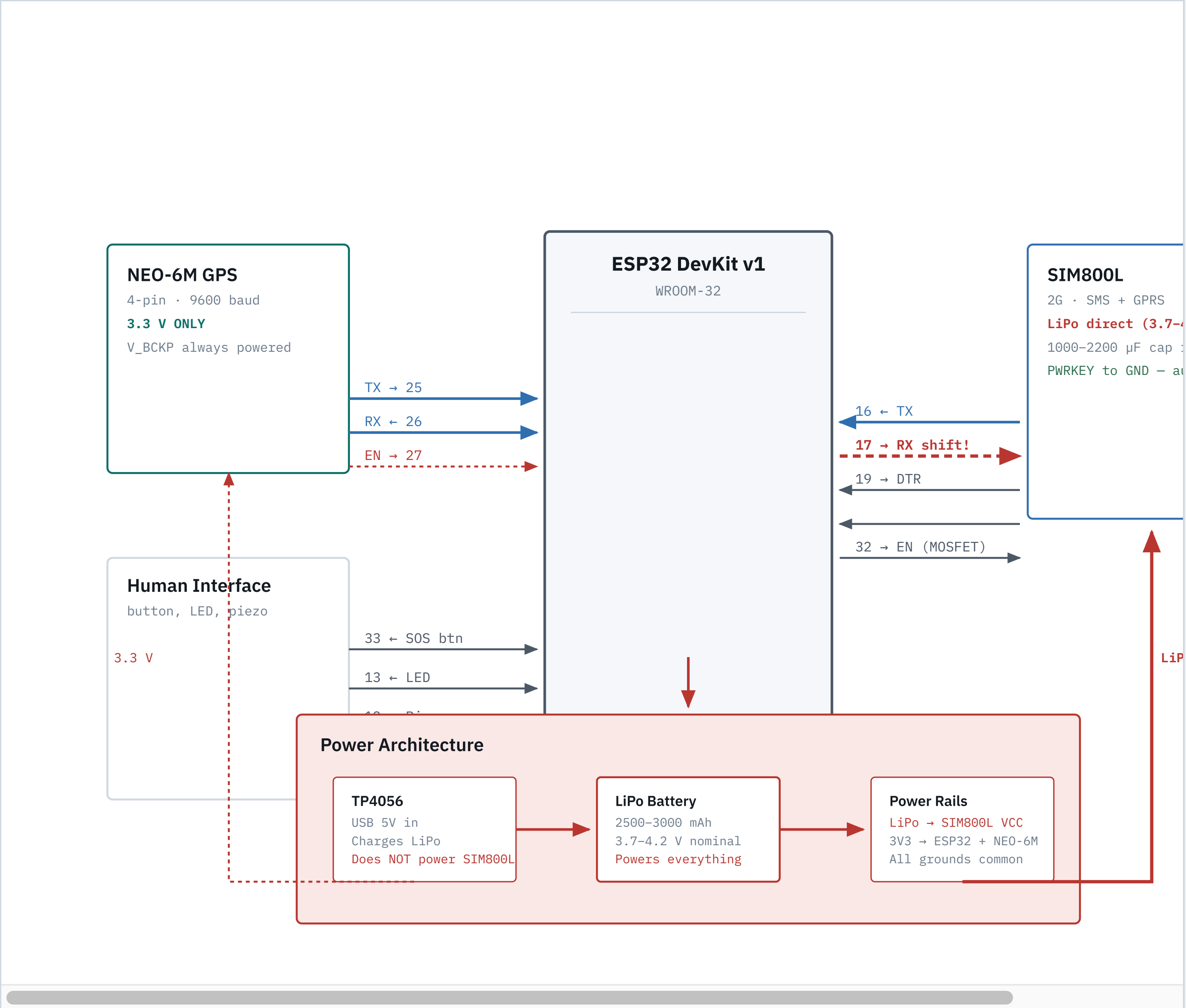
The SIM800L draws up to 2 A in transmit bursts at 3.4–4.4 V. It cannot run from the ESP32's 3V3 pin, from the DevKit's 5V pin, or from the TP4056 output. Wire it **directly to the LiPo battery**, which sits happily in that voltage range.

Fit a **1000–2200 µF electrolytic across the SIM800L's VCC/GND, physically close to the module**. Without it the TX burst browns out the rail, the ESP32 resets, and you spend a weekend blaming the firmware.

The NEO-6M is 3.3 V only. 5 V kills it. The SIM800L RX is not 3.3 V tolerant — level-shift the ESP32 TX line.

01 Connection diagram

UART1 — GPS   UART2 — SIM800L   Power   GPIO / control   Needs level shifting



The SIM800L has its own power path directly from the LiPo. The TP4056 charges the battery but does not power the modem — the battery is the supply. The 3.3 V rail feeds only the ESP32 and NEO-6M. All grounds must be tied together.

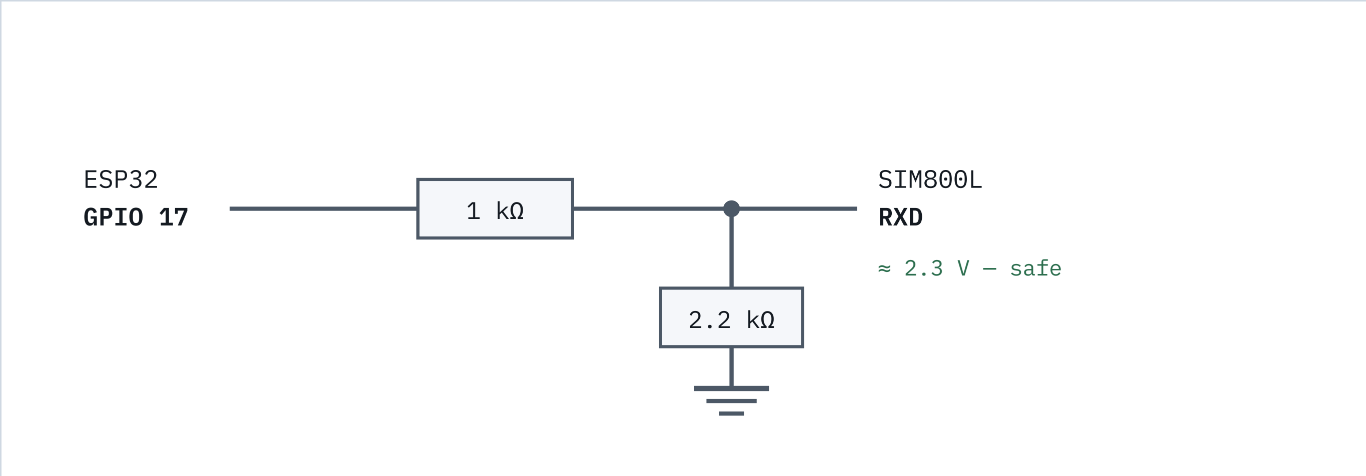
02 Connection table

MODULE	MODULE PIN	ESP32	BUS	NOTE
NEO-6M	TX	GPIO 25	UART1	GPS data → ESP32. 9600 baud
NEO-6M	RX	GPIO 26	UART1	ESP32 → GPS config
NEO-6M	VCC / GND	3V3 / GND	PWR	**3.3 V only. 5 V kills it**

MODULE	MODULE PIN	ESP32	BUS	NOTE
NEO-6M	EN (MOSFET)	GPIO 27	GPIO	Power gate. V_BCKP stays powered always
SIM800L	TXD	GPIO 16	UART2	~2.8 V out reads as HIGH; no shifting needed
SIM800L	RXD	GPIO 17	UART2	**Level-shift.** RX is not 3.3 V tolerant
SIM800L	PWRKEY	–	PWR	Tied to GND internally — module auto-powers on when VCC is applied
SIM800L	DTR	GPIO 19	GPIO	CSCLK sleep: HIGH = sleep, LOW = wake
SIM800L	EN (MOSFET)	GPIO 32	GPIO	High-side MOSFET gate, full power cut
SIM800L	VCC / GND	LiPo / GND	PWR	**Direct to LiPo (3.7–4.2 V). Own supply**
SOS Button	one leg	GPIO 33	GPIO	Active LOW, internal pull-up, RTC-capable
SOS Button	other leg	GND	PWR	Direct to ground
LED	anode	GPIO 13	GPIO	220 Ω series to GND
LED	cathode	GND	PWR	Through 220 Ω resistor
Piezo	signal	GPIO 12	GPIO	Passive buzzer, tone() at 2400 Hz
Piezo	GND	GND	PWR	
Onboard LED	—	GPIO 2	GPIO	Debug only — not the child-facing one

03 Level shifting GPIO 17

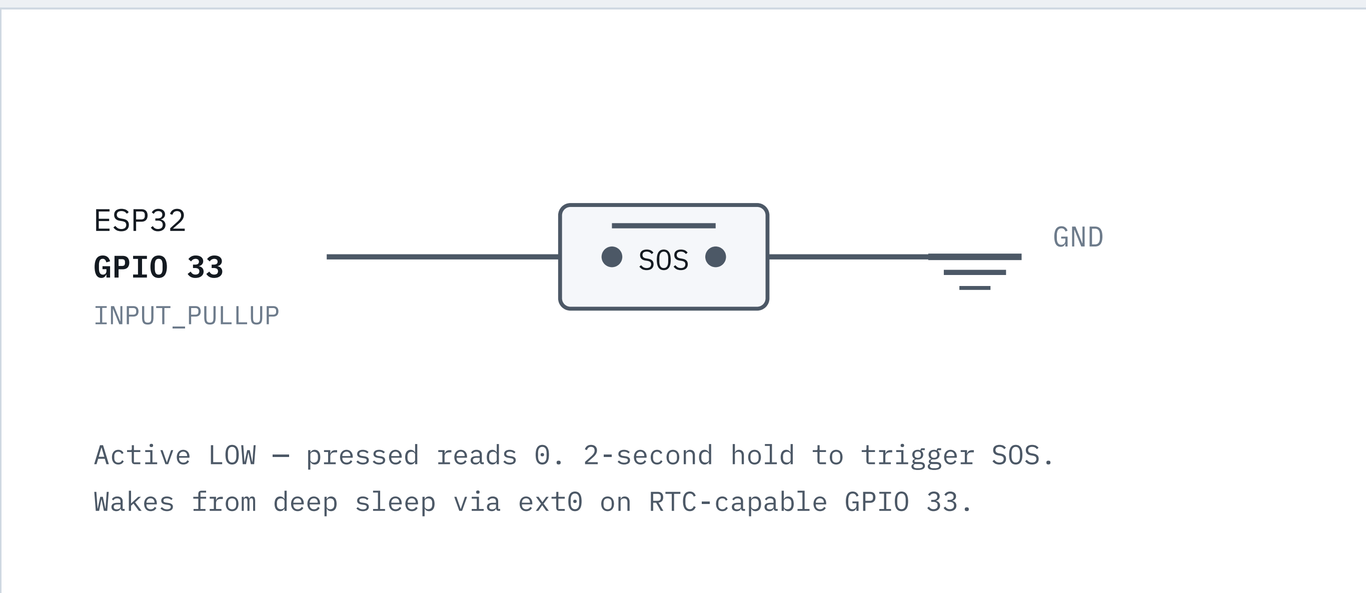
Only one line needs it. The SIM800L's RX input is not 3.3 V tolerant, and driving it directly works on the bench for a while and then degrades the pin. A two-resistor divider is enough for a one-way 9600-baud line.



3.3 V × 2.2/(1+2.2) ≈ 2.3 V. The other direction needs nothing: the SIM800L's ~2.8 V output already reads as a logic HIGH on the ESP32.

04 SOS button

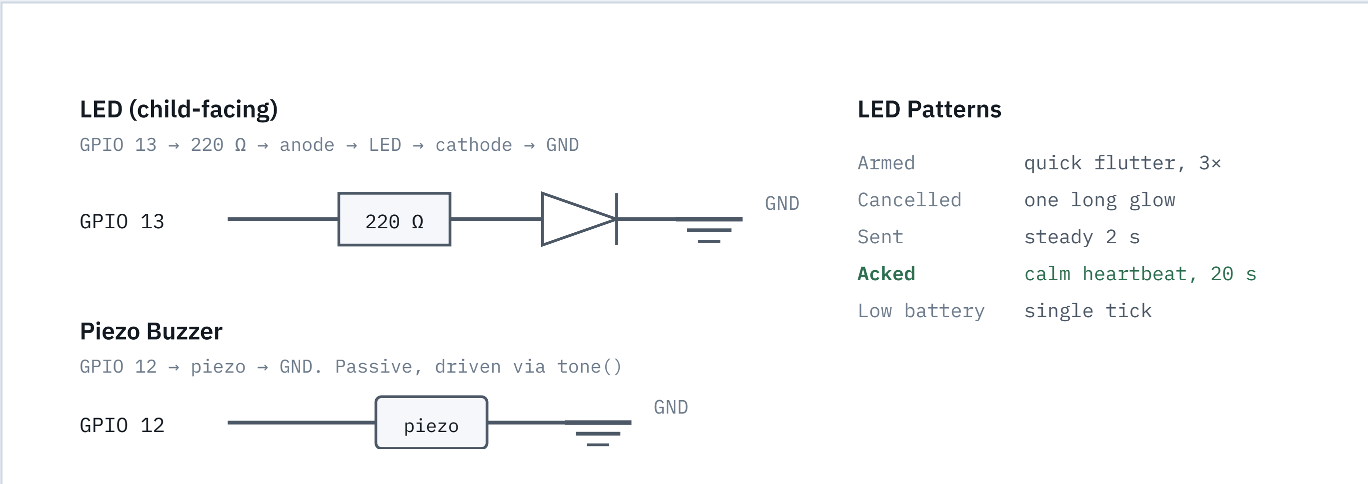
A recessed momentary tactile button between GPIO 33 and GND. The ESP32's internal pull-up holds the line HIGH; pressing the button pulls it LOW. GPIO 33 is RTC-capable, so it can wake the ESP32 from deep sleep via `ext0`.



Why GPIO 33: Deep-sleep wake requires an RTC-capable GPIO. GPIO 33 is one of the few. The 2-second hold is enforced in firmware to prevent pocket false alarms.

05 LED + Piezo

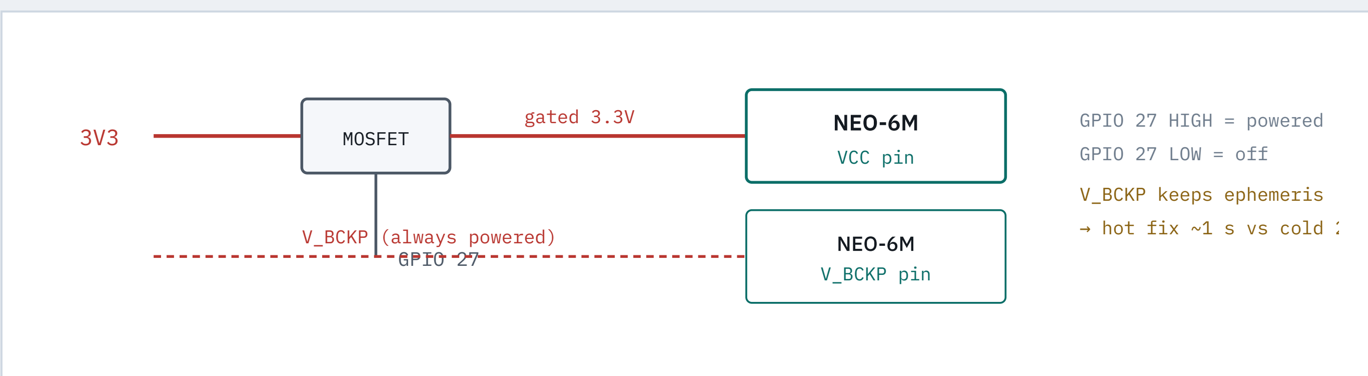
The child-facing feedback. An LED on GPIO 13 and a passive piezo buzzer on GPIO 12. The LED stands in for a vibration motor (out of stock) — the motor will drop onto the same GPIO later with a MOSFET and flyback diode.



Mount the LED where the child can see it — a bezel or light pipe on the front face, not buried in the enclosure. If you have a piezo, use it: audible cues reach a child who is not looking at the device.

06 NEO-6M power gating

GPIO 27 drives a high-side MOSFET that gates 3.3 V to the NEO-6M's VCC pin. The critical detail: **V_BCKP must stay powered regardless**. If you cut backup power, every GPS fix becomes a 27-second cold start instead of a ~1-second hot fix — and the cold start costs more energy than the gating saved.



Power-gating the NEO-6M saves ~45 mA between fixes. But never cut V_BCKP — the ephemeris cache is what makes a hot fix fast, and rebuilding it costs more energy than the gating saved.

07 Bring-up, one module at a time

Wiring everything then flashing the full firmware means any fault presents as "nothing works". Bring them up in this order — each step is independently verifiable.

`01` **ESP32 alone**

Blink GPIO 2 (onboard), GPIO 13 (LED), and tone GPIO 12 (piezo). Confirms the board, the supply, and your feedback wiring before anything else can confuse the picture.

`02` **NEO-6M**

Power from 3V3. Open a serial monitor on GPIO 25/26 at 9600 baud. You should see NMEA sentences within a minute outdoors, five indoors. If all zeros or no data: TX/RX swapped, or module is 5 V dead.

`03` **SIM800L on its own supply**

Before any code: power from LiPo, watch the status LED blink, then `AT` → `OK`, `AT+CREG?` → `0,1` or `0,5`, `AT+CSQ` ≥ 10. Send one SMS by hand with `AT+CMGF=1` and `AT+CMGS`. This breakout ties PWRKEY to GND — it powers on as soon as VCC is applied.

`04` **SOS button**

Upload a sketch that prints `digitalRead(33)` to serial. Press the button — should go LOW. If it never reads LOW: wiring fault or wrong GPIO. If it reads LOW without pressing: short circuit.

`05` **Full firmware**

Only now. Serial should print `tracker up` with modem and GPS status. If GPS shows "no fix" indoors, that is normal — take it outside.

08 Symptom → cause

SYMPTOM	ALMOST ALWAYS
ESP32 reboots when SMS is sent	**Missing or undersized bulk cap** on the SIM800L rail. The single most common fault on these builds
SIM800L never answers `AT`	TX/RX swapped. Try swapping them — it costs nothing
Answers `AT`, garbled replies	Baud mismatch, or GPIO 17 driven without the level shifter
`AT+CREG?` returns `0,2` forever	Searching: no 2G coverage, SIM not registered under RA 11934, or load expired

SYMPTOM	ALMOST ALWAYS
GPS cold start every fix (27 s)	V_BCKP not powered — ephemeris lost between power cycles
GPS no data at all	TX/RX swapped, or NEO-6M powered from 5 V (likely dead)
SOS button does not wake from deep sleep	Wrong GPIO — must be RTC-capable (GPIO 33). Check `ext0` config
LED always on or inverted	Not applicable for single LED — check wiring direction if reversed
Everything works on USB, fails on battery	LiPo voltage sag under 2 A TX burst. Use a fresh cell, check connections

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Matches `tracker/include/pins.h` at the time of writing — if you change a pin, change it there first and regenerate this. GPIO 12 sets flash voltage at boot — it is not safe for arbitrary reuse.